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# A landscape of bioinformatics patents - Garnering of IPR in the field of bioinformatics

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ABSTRACT

In the current information technology era, Bioinformatics is growing rapidly due to availability of vast database systems and the ever increasing amount of biological data. It is a flexible and creative means of storing, managing, and querying of complex biological datasets. With these rapid advancements in today's technology-driven age, it is also imperative that protection in the form of Intellectual Property Rights (IPR) is sought for such research and development activity. In addition, there is a need to formulate an aggressive strategy to protect one's IP. In fact, various companies', universities', institutions' and researchers' are into the process to protect their core invention. This landscape will give an outline of the latest technological growth, geographical distribution, and top competitors playing an important role in this field.

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**Contents**



*Abbreviations:* IPR, Intellectual Property Right; R&D, Research and Development; IPC, International Patent Classification; CPC, Cooperative Patent Classification; IPR, Intellectual Property Rights; SB, Modelling and Simulations in Systems Biology; PE, Phylogeny or Evolution; MS, Molecular Structure; HE, Hybridization or Gene Expression; SC, Sequence Comparison; MDB, Machine Learning, Data Mining or Biostatistics; PD, Programming tools or Database Systems; DV, Data Visualization; GP, Functional Genomics or Proteomics.

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## 1. Introduction

Bioinformatics is a merger of biotechnology and information technology, which directs the use of biostatistics, mathematics, algorithms, programming tools and databases [1]. It comes with an ability to organize, analyze and store most of the biological data, generated through high-throughput experiments [2]. One of the primary aims of bioinformatics field is to help in reducing the time, money and efforts in the process of drug design and discovery [3]. It also plays crucial role in analyzing large chunks of biological sequence data and to predict the possible functions of genes and proteins [2]. The increase in the information and data from various fields in biotechnology such as genomics, proteomics and transcriptomics have resulted in development of more advanced bioinformatic tools for the analysis of specific data. Researchers developing bioinformatic tools are now more concerned about protecting their inventions in the form of patent rights [4].

## 2. Background

Currently almost for every biological research field, bioinformatics has not only reduced the long span of experimental procedures to short but also has provided lots of tools and databases where the experimental data is kept and has benefited biological researchers world wide. Nowadays, it is possible to perform highly accurate searches using software systems and leading bioinformatic companies are working on such developments with an aim to provide benefit to whole area of biology. Various analytical tools, software and hardware products generated from bioinformatics have supported biotechnology and pharmaceutical industry to a great extent. There is a huge amount of investment and risk involved in R&D for performing bioinformatic related inventions [15] , hence for such R&D activities, patent protection is a necessary. Patent protection also helps companies to gain returns on the investments by licensing out the technologies [16].

Patents are granted by national governments for those innovations, which are novel, not obvious and have an industrial application. After the grant, the applicants get a right to prevent others from using the invention for limited period of time (20 years). A patent application may take upto 18 months to get published after the date of filing, so, there is a time lag in patent procedure between the filing and publication of the patent application.

The patent applications are assigned a classification code according to International Patent Classification by patent office, based on the technical features of the patent application. This International Patent Classification (IPC) system is accepted by most of the patent granting authorities. In recent years, major patent granting authorities have started using the Cooperate Patent Classification (CPC) system for classification. CPC, an extended version of IPC, includes more specific details of the technical features. Both these patent classification systems, are well-accepted. According to IPC and CPC systems, bioinformatics is broadly classified in the following ways;

### 2.1. Modelling or simulation in systems biology (SB)

Systems Biology involves the computer simulations or mathematical modelling to study the complex interactions between the biological systems. It can be considered for various aspects like studying the interactions between biological components, their dynamic behavior under varying conditions, and to predict their properties [8].

### 2.2. Phylogeny or evolution (PE)

Phylogeny is classification of individuals or group of organisms by their evolutionary history. So, phylogenetics is referred to the taxonomical classification and represented with the development of the phylogenetic tree [9].

### 2.3. Molecular structure (MS)

In the field of structural biology, bioinformatics helps in predicting the properties and functions of biological macromolecules such as proteins, and nucleic acids (RNA and DNA). It also covers the alterations of structures and an effect on their functioning [10].

### 2.4. Functional genomics or proteomics (GP)

With the advancements, this branch of bioinformatics uses the computational and high-throughput experiments for studying of genes and proteins, and focuses on the dynamic aspects such as gene transcription, translation, and protein-protein interactions [11].

### 2.5. Hybridization or gene expression (HE)

The analysis of gene expression can be performed via measuring the mRNA levels. It includes microarray analysis, gel electrophoresis, and sequence hybridization [12].

### 2.6. Sequence comparison (SC)

Sequence comparison is the most significant field in bioinformatics. It involves studying the sequences of DNA, RNA, and proteins, and identifies the functional, structural, or evolutionary relationships between the sequences [13].

**Table 1**
Description of Classification System. The list of International and Cooperative Patent Classification System is shown below.

| S.no. | IPC or CPC | Definition | Description |
|---|---|---|---|
| 1 | G06F 19/12 | Modelling or simulations in systems biology | Simulation or mathematical modelling of relationships and interactions between molecular entities on a subcellular level<br>Integrating genetic and/or protein-related data to describe the dynamic behavior of protein-protein/protein-ligand interactions<br>Regulatory or metabolic networks |
| 2 | G06F 19/14 | Phylogeny or evolution | Analysis of orthologous, paralogous, syntenic or taxonomic relationships.<br>Generation of pedigrees and phylogenetic trees. |
| 3 | G06F 19/16 | Molecular structure | Structural architecture of proteins, peptides, amino acids and nucleic acids and the prediction thereof; The structure types include secondary, tertiary and quaternary structures.<br>Processes including structural alignment, protein folding, domain topology, molecular modelling, receptor-ligand modelling, docking methods, structural-functional relationships and drug targeting using structure data, as well as two- and three-dimensional structure prediction and/or analysis. |
| 4 | G06F 19/18 | Functional genomics or proteomics | Assessment of the function of genes and proteins in determining traits<br>Physiology and/or development of an organism,<br>Making use of computational and large scale, high-throughput technologies.<br>Genotypic-phenotypic associations, including genotyping and genome annotation, linkage disequilibrium analysis and association studies, population genetics, alternative splicing and Short Interfering RNA design (siRNA, RNAi).<br>Binding site identification, mutagenesis analysis, protein-protein or protein-nucleic acid interactions. |
| 5 | G06F 19/20 | Hybridization or gene expression | Analysis of gene expression information; includes microarray analysis, gel electrophoresis analysis and sequencing by hybridization.<br>Further covered technologies include probe design and probe optimization, microarray normalization, expression profiling, and noise correction models, expression ratio estimation. |
| 6 | G06F 19/22 | Sequence comparison | Comparison of sequence information, wherein the sequences are nucleic acids or amino acids include; methods of alignment, homology identification, motif identification, SNP (Single-Nucleotide Polymorphism) discovery, haplotype identification, fragment assembly, gene finding. |
| 7 | G06F 19/24 | Machine learning, data mining or biostatistics | Discovery and/or analysis of patterns within a vast amount of genetic or protein-related data, wherein the emphasis is placed on the method of analysis and is largely independent of the type of bioinformatic data.<br>Covered methods include; bioinformatic pattern finding, knowledge discovery, rule extraction, correlation, clustering and classification and Multivariate analysis of protein or gene-related data. |
| 8 | G06F 19/26 | Data visualization | Visualization of bioinformatic data specifically includes; graphics generation, map display and network display. |
| 9 | G06F 19/28 | Programming tools or database system | Computer software specifically adapted to assist programming procedures within bioinformatics and database systems specifically adapted for managing bioinformatic data, includes; ontologies, heterogeneous data integration, and data warehousing, and computing architectures. |

### 2.7. Machine learning, data mining or biostatistics (MDB)

This technical area explores genes or proteins related data, disease diagnosis and their treatment using numerous prediction models and methods that also include statistical analysis. Besides, it also helps in creating meaningful patterns, correlations, and clustering of the vast amount of data. It finds a significant role in the extraction of knowledge from such biological datasets [14].

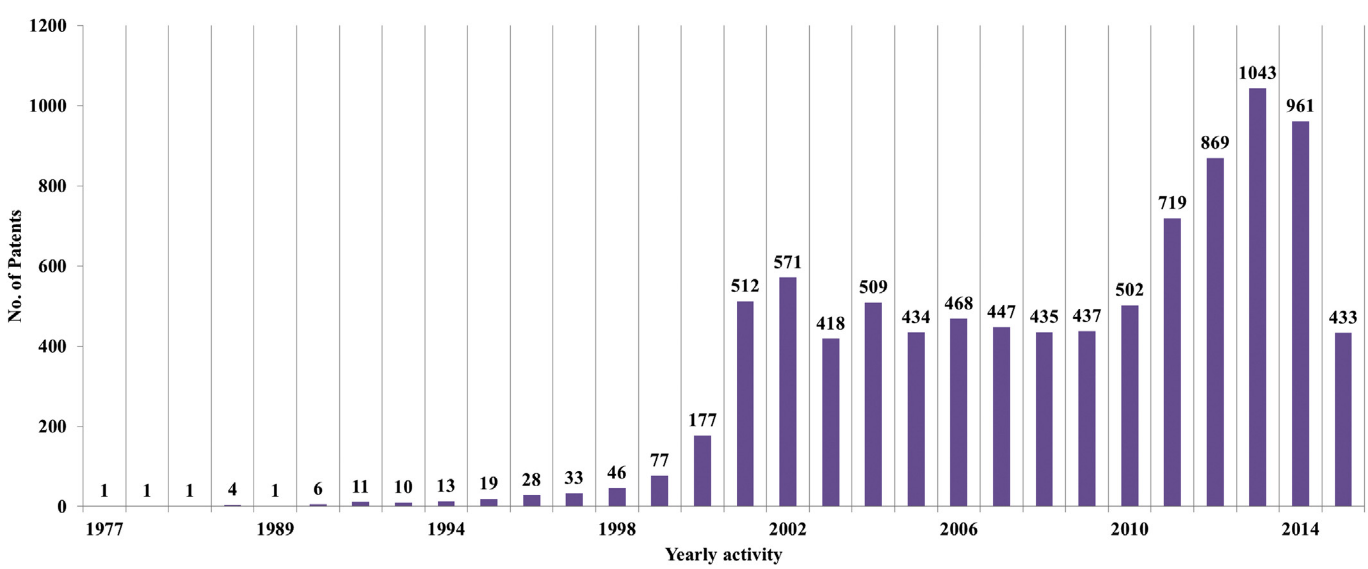


**Fig. 1.** Patent Filing Trend. The year wise patenting activity in relation to bioinformatics related inventions.

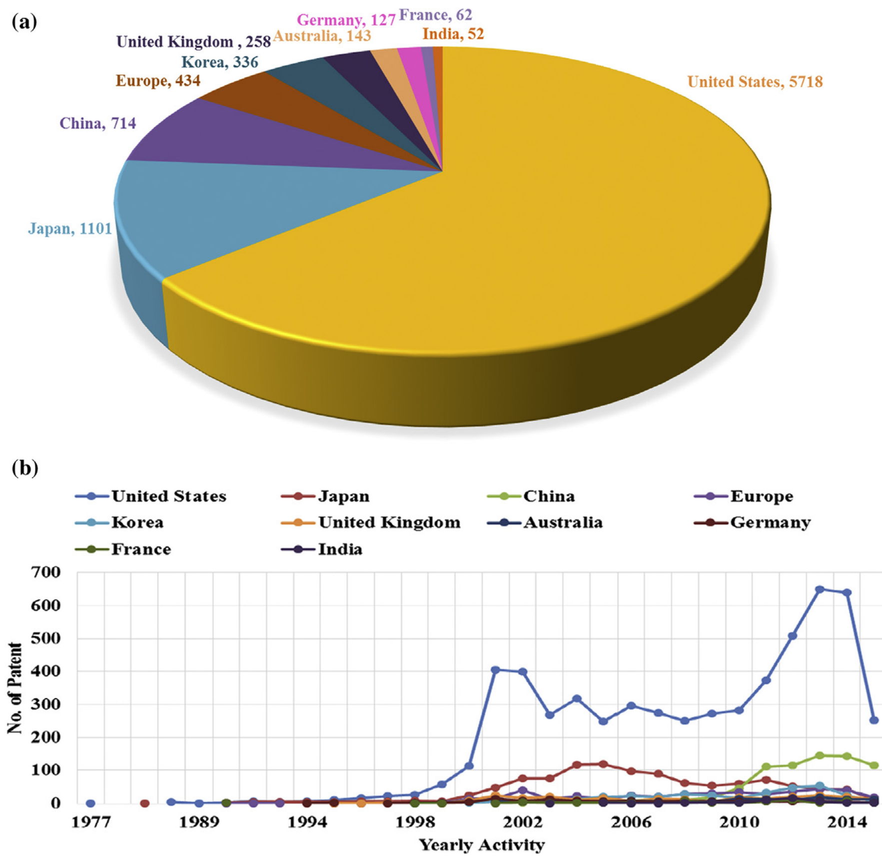


**Fig. 2.** (a). Geographical distribution. The priority country-wise patent filing trend. (b). Country-wise distribution. The trend represented with respect to year wise patent/application filings.

*2.8. Data visualization (DV)*

This group of technology is particularly used for the representation of biological datasets and information extracted from them. It includes computer graphics, tools for visualizing the structure of biological macromolecules, networks representation, and sequence visualization [15].

*2.9. Programming tools or database systems (PD)*

Due to advancements in high-throughput techniques, there is a huge availability of data, and numerous biological databases have been developed for storage and communication purposes. The application of this area of bioinformatics are storing data for future usage that includes sequences, structural and gene expression data, numerous metabolic pathways, phylogenetics trees and various other type of information related to biological function and processes [16].

The details of above mentioned nine technologies and the concerned IPC and CPC classification codes are further elaborated in Table 1.

A patent is a shield for one's creation of mind and thus, considered as the strongest indicators in representing the technical value of organizations [17], and the information from patents can be used in technology monitoring, observing the competitive environment, strategic management and decision-making in business community [18].

## 3. Materials and methods

In the current information technology era, use and need of bioinformatics is ever increasing [19]. In this study, our focus was to analyze the technological growth, trends and major active players who were involved in the patenting activity.

The classification of the patent was carried out using Cooperative Patent Classification (CPC) and International Patent Classification (IPC) system and categorizing bioinformatic technologies into nine broad classes (Table 1).

A search strategy was prepared which included specific keywords and classification codes to recover patents available till December 2015 from Thomson Innovation database. In search strategy we used the CPCs/IPCs of all the 9 technologies along with bioinfo* OR comput ADJ Bio* as key words. A total number of 10,208 patents/applications were retrieved (INPADOC Patent Families), of which 1021 patents/applications were not related to bioinformatics area. Further, the patents/applications were classified into nine broad technological areas. The assignees for patents/applications were standardized using the corporate tree feature of Thomson Innovation. For example, Life Technologies Inc. was acquired by the Thermo Fisher Scientific Inc. in 2003. So, all the patents belonging to Life Technologies Inc. were classified under Thermo Fisher Scientific Inc. The technologies, assignees involved and other trends were analyzed/visualised using VantagePoint 9.0 software.

## 4. Result and discussion

Information from bioinformatics patents was analyzed based on the classified technologies. The visualisation is represented by various charts that reveals the patent filing trends, technology growth, top assignees, and assignee-wise IP activity for the classified technologies.

### 4.1. Technology growth

The patent filings in this technology area began in 1970's and steadily grew for four decades as depicted in Fig. 1. The technology growth started in the year 1999 and reached peak in year 2013.

The geographical distribution indicates that United States (US) is the worldwide leader in the Bioinformatics domain followed by Japan and China as revealed from Fig. 2. Fig. 2 also shows that inventors from United States began filing from the year 1977 and reached a peak in the year 2013. Similarly, patent fillings by Japan was started from 1986 and peaked in the year 2004. China was a late entrant into this field (2002) with maximum fillings in 2013 and they have overtaken Japan.

### 4.2. Technical categorization

Various technologies associated with bioinformatics are shown in Fig. 3. The analysis revealed that Functional genomics or proteomics (GP) is leading technology followed by Sequence comparison (SC), Modelling and Simulations for Systems Biology (SB) and so on with respect to the filing activity of patents/applications. Although, the patenting activity was observed from 1977, however, they came into prominence after 1990s as shown in Fig. 3. It was observed that amongst nine leading technical areas in the field of Bioinformatics, the research and patenting activity in the Functional genomics or proteomics (GP) was the most sought after. An increase in patent/application filings in the years 2013 and 2014 was observed for the leading technologies related to functional genomics or proteomics, sequence comparison (SC) and Machine Learning, Data Mining or Biostatistics (MDB). The details of other assignees and their chronological growth for respective technologies had been discussed in later sections.

### 4.3. Major assignee

For companies, innovation is the most important means of staying competitive in the marketplace, and the only way is to protect innovative ideas and prevent other companies from using them. Patents provide a way to exclude others from using your idea, at least for a limited period [21]. The assignee-based analysis for bioinformatics patents under the study was carried out to identify the top active players over a period. The top three applicants' viz. Thermo Fisher Scientific Inc, Hitachi Ltd and IBM Corp were involved in the patenting of multiple technologies related to Bioinformatics area as showed in Fig. 4. It was observed that apart from these three top players, many other organizations are also setting up their roots in this area. The timeline in Fig. 4 reveals the patent filing activity of the top 10 assignees involved in nine broadly classified technologies. It was observed that patenting activity of Thermo Fisher Scientific Inc. increased after the year 2012. Whereas, Hitachi Ltd. was the active and dominant player only in the years 2002–2004. Moreover, the assignees like BGI Shenzhen Co Ltd, IBM Corp, Philips N.V., and Roche Holding Ltd. have increased their patenting activity significantly after year 2011.

### 4.4. Assignee-wide distribution of Patent activity

Overall assignee-wise distribution of patenting activity (Fig. 5)

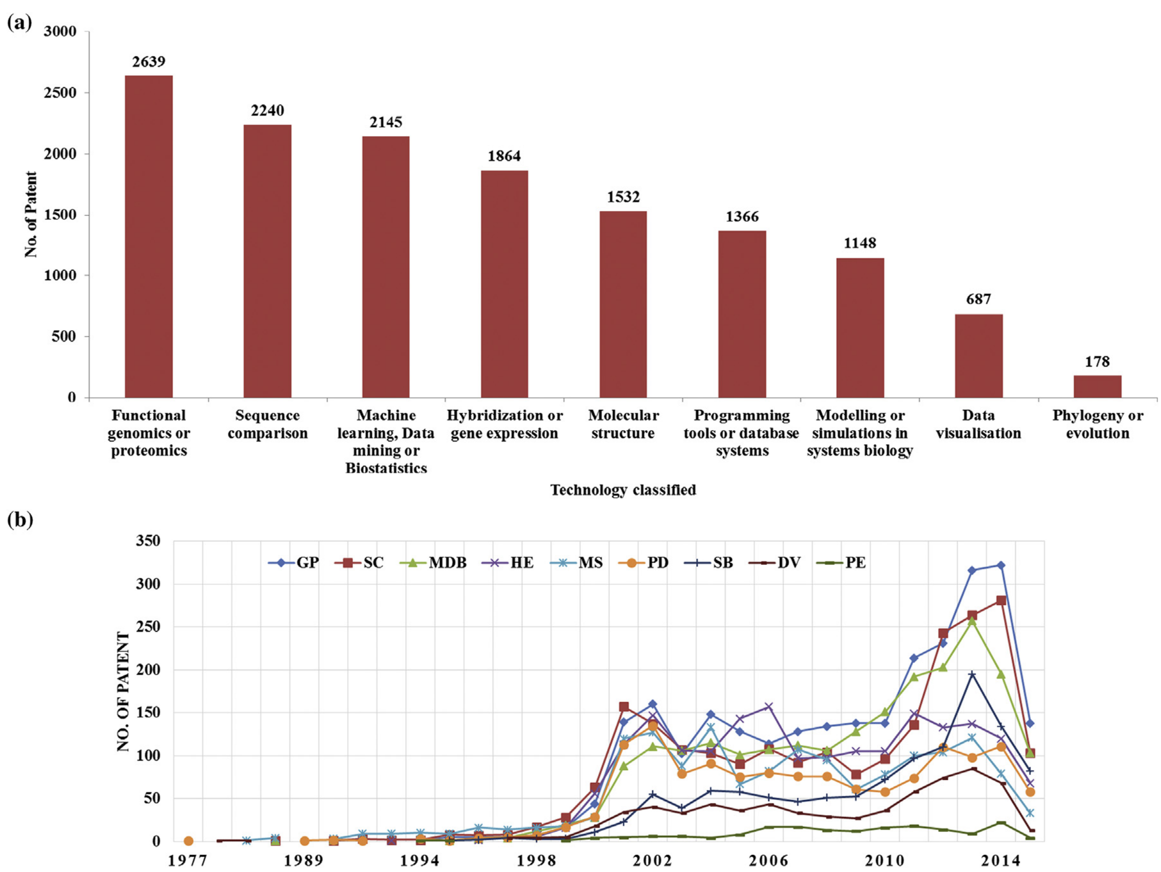


**Fig. 3.** (a). Technology Classification. The count of number patents/applications fall under nine broadly classified technologies. (b). Technology wise patenting trend. It reflects the patenting activity with respect to the filing of patents/applications.

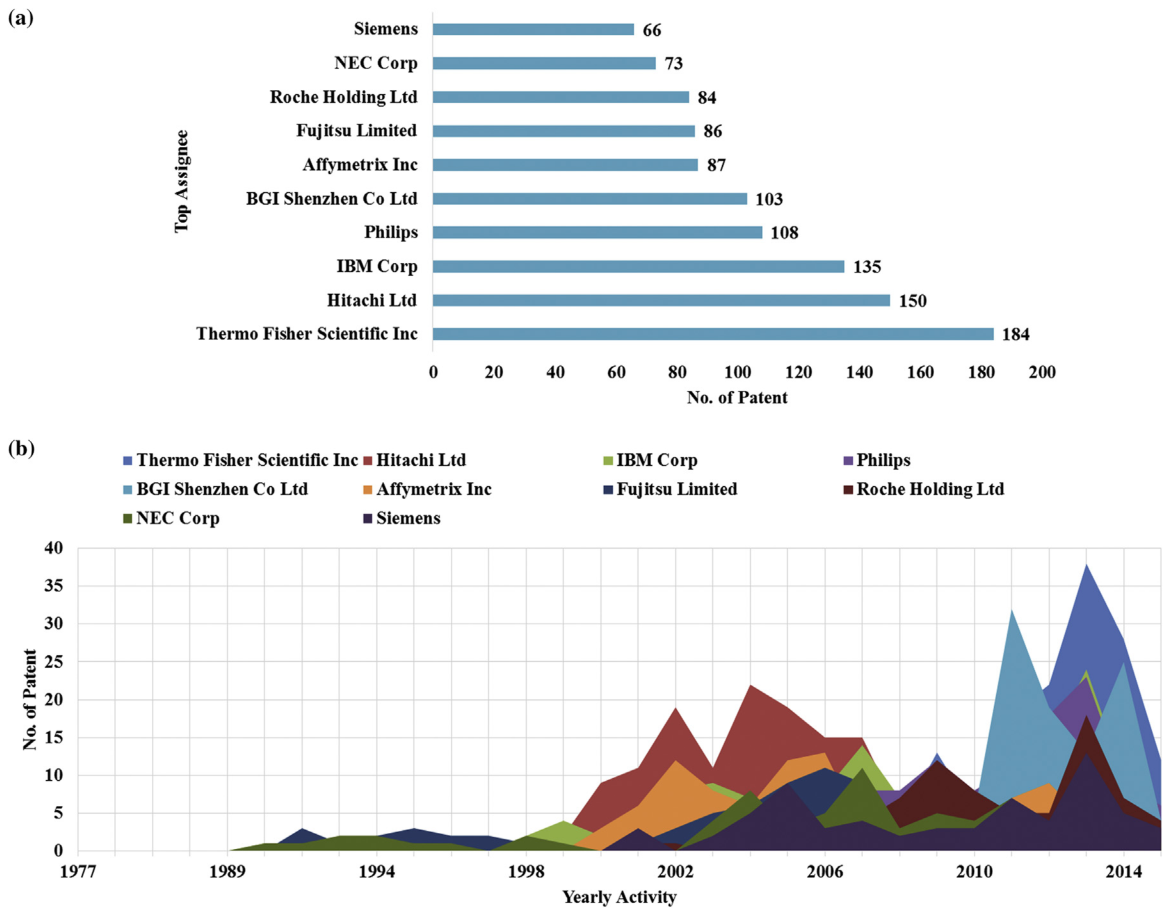


**Fig. 4.** (a). Top Assignees. It shows the top-10 active players along with the count of patents/applications. (b). Assignee-wise patent filing trend. The year wise patenting trend by top active players in the domain.

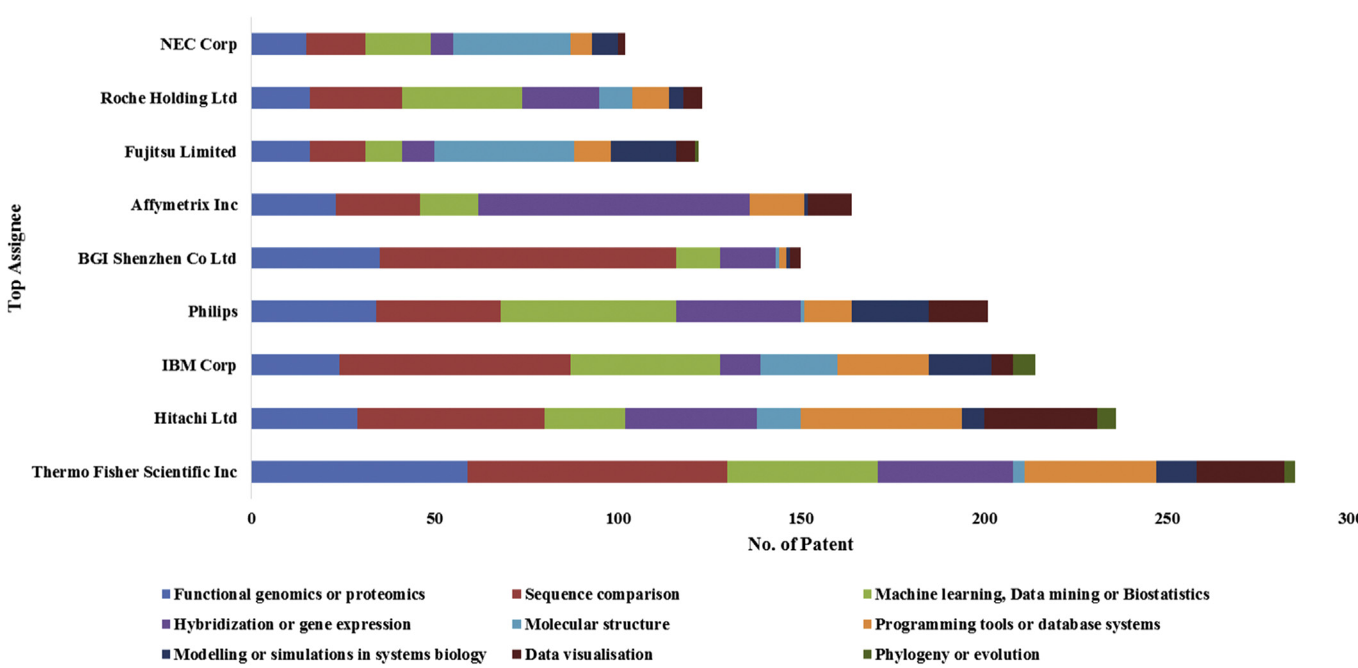


**Fig. 5.** Assignee-wide IP activity. It represents the distribution with respect to number of patents/applications filed by top assignee into nine classified technologies.

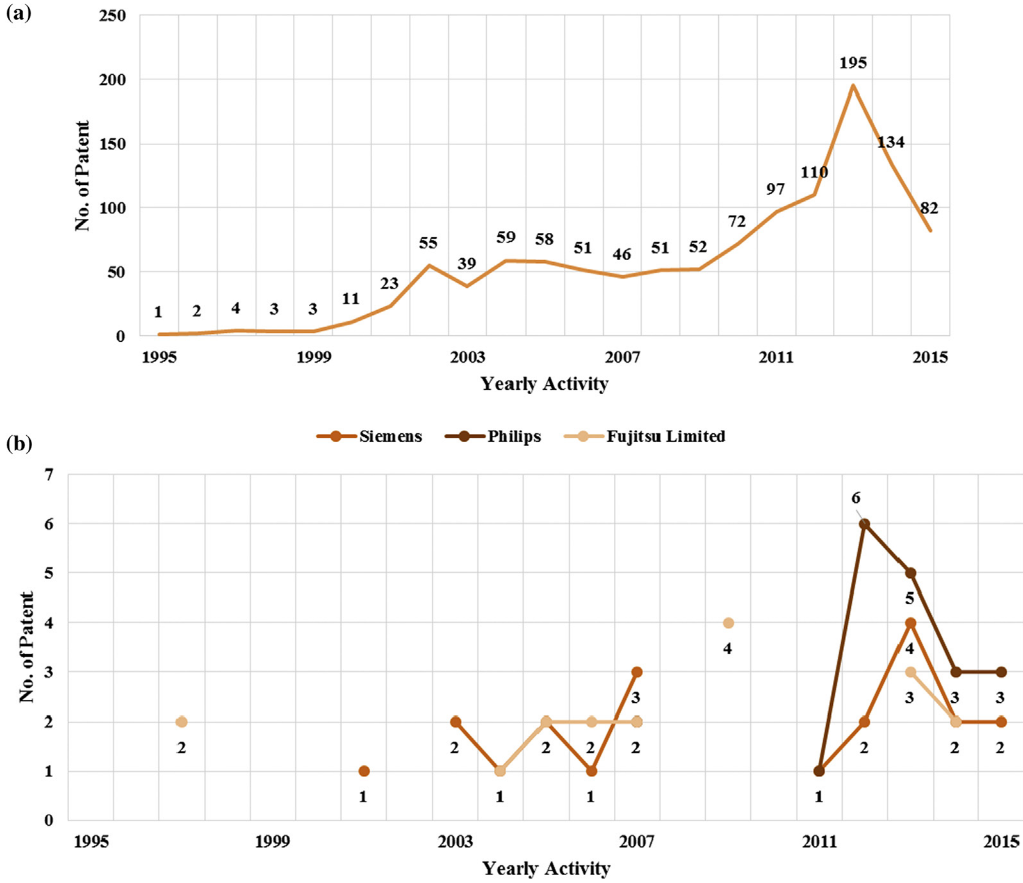


**Fig. 6.** (a). Patenting Trend in Modelling or Simulation in Systems Biology (SB). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Modelling or Simulation in Systems Biology (SB).

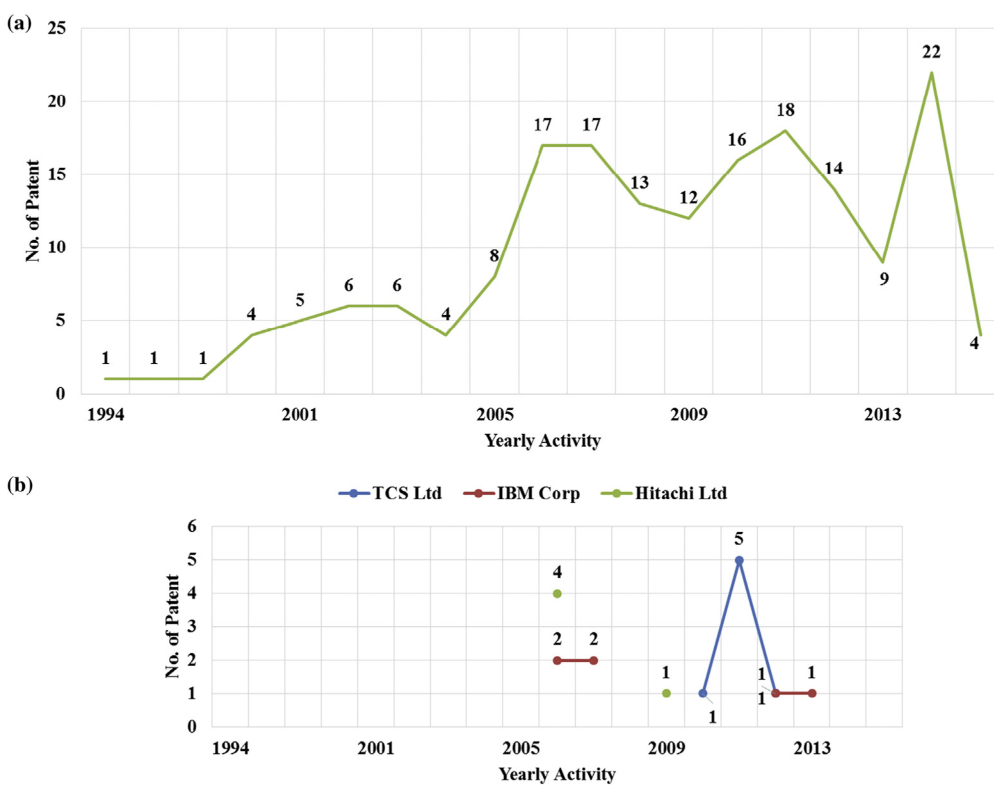


**Fig. 7.** (a). Patenting Trend in Phylogeny or Evolution (PE). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Phylogeny or Evolution (PE).

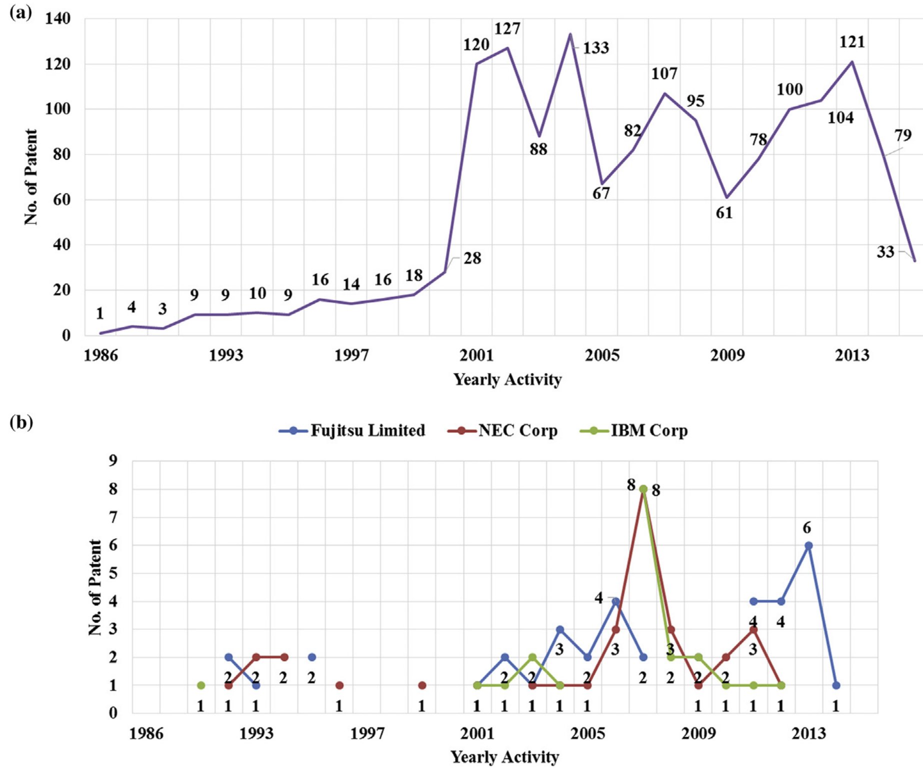


**Fig. 8.** (a). Patenting Trend in Molecular Structure (MS). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Molecular Structure (MS).

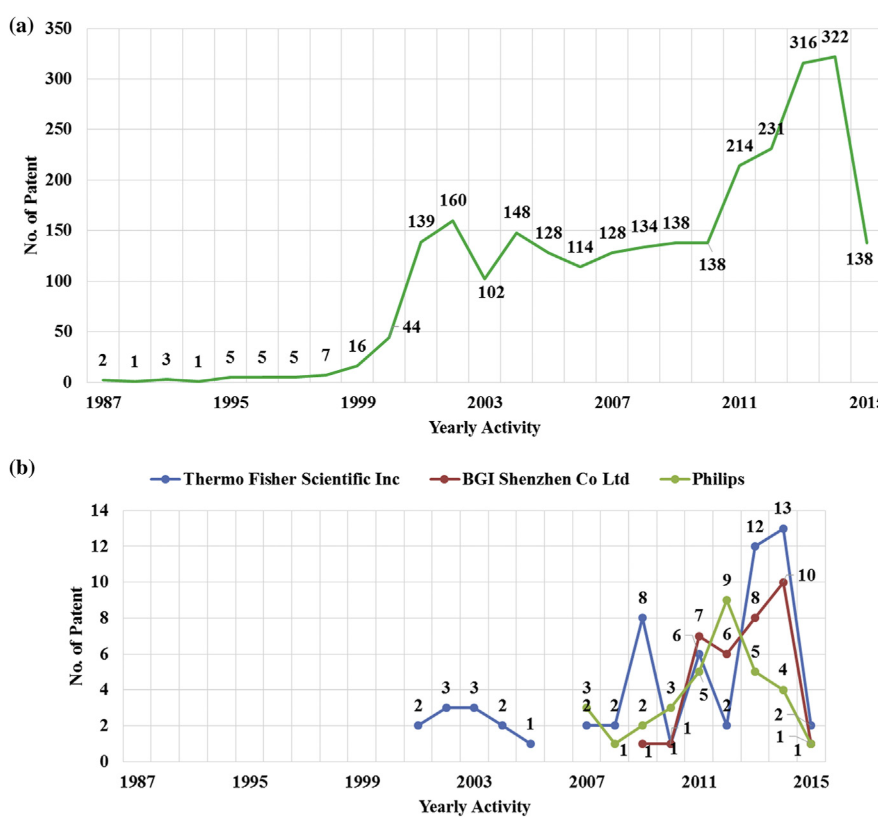


**Fig. 9.** (a). Patenting Trend in Functional Genomics or Proteomics (GP). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Functional Genomics or Proteomics (GP).

showed that Thermo Fisher Scientific Inc. and BGI Shenzhen Co Ltd were involved in Sequence comparison (SC), Functional Genomics or Proteomics (GP), and Hybridization or Gene Expression (HE). Likewise, Hitachi Ltd and Philips N.V. were mostly involved in Modelling or Simulation in Systems Biology (MDB), Sequence Comparison (SC), and Hybridization or Gene Expression (HE) whereas Affymetrix was the top filer fot Hybridization or Gene expression (HE) related patents.

### *4.4.1. Modelling or simulation in systems biology (G06F 19/12)*

Modelling or simulation is a technical area in bioinformatics which helps in analyzing the processes in the cell with the usage of computer simulations and mathematical models [6]. Fig. 6 represents the gradual increment of patents/applications filings and the activity peaked in the years 2010–2013 but again fell in later years. Fig. 6 also reveals that Siemens and Philips N.V. had almost similar numbers of patents/applications filed in years 2012–2013 and 2011–2013, respectively, followed by Fujitsu Ltd.

### *4.4.2. Phylogeny or evolution (G06F 19/14)*

Phylogenesis involves the taxonomical classification of organisms by developing pedigrees and phylogenetic trees [7]. Fig. 7 reflects the movement of patents/applications based on filing activity . Fig. 7 shows almost similar number of patents/applications filings with respect to time. TCS has raised their number in the year 2011, focused specifically on phylogeny or evolution domain.

### *4.4.3. Molecular structure (G06F 19/16)*

The structure prediction of biological macromolecules majorly proteins, RNA, and DNA are covered in the field of molecular structure [8]. Fig. 8 displays highest filings in the years 2002, 2004 & 2013. According to Fig. 8. Fujitsu Ltd. did not file for many years (1995–2000) however, its filings peaked up in 2012. On the other hand, NEC Corp and IBM Corp had most of the patents/applications filings between 2005–2008 but still lagged behind Fujitsu in overall filings.

### *4.4.4. Functional genomics or proteomics (G06F 19/18)*

It includes large-scale analysis of genes and proteins expression [20]. It is revealed from Fig. 9 that the filing activity was constant till 1998 and gradually increased between the years 1998–2002 and the technology peaked up in the years 2013 and 2014. Fig. 9 reveals that Thermo Fisher Scientific Inc. was active in recent years as compared to other two assignees.

### *4.4.5. Hybridization or gene expression (G06F 19/20)*

It builds on molecular biological research and includes analysis of gene expression [10]. Fig. 10 explains start-up of patent protection was observed from the 1990s. Fig. 10 reveals that Affymetrix Inc. is the major player in the area and their patenting activity was maximum in years 2001–2006. However, filings from Thermo Fisher Scientific Inc. peaked up in the year 2013 along with Affymetrix Inc.

### *4.4.6. Sequence comparison (G06F 19/22)*

It has the major role in the data integration, which organizes and analyzes the biological data [11]. Fig. 11 shows that patents/applications filing were started after the year 1985. Fig. 11 represents the race between the leading assignees where their patent filing number was almost similar in the years 2010–2015. It was noticed that the BGI Shenzhen Co Ltd. held the maximum number of patents/applications with its activity at peak in the year 2011.

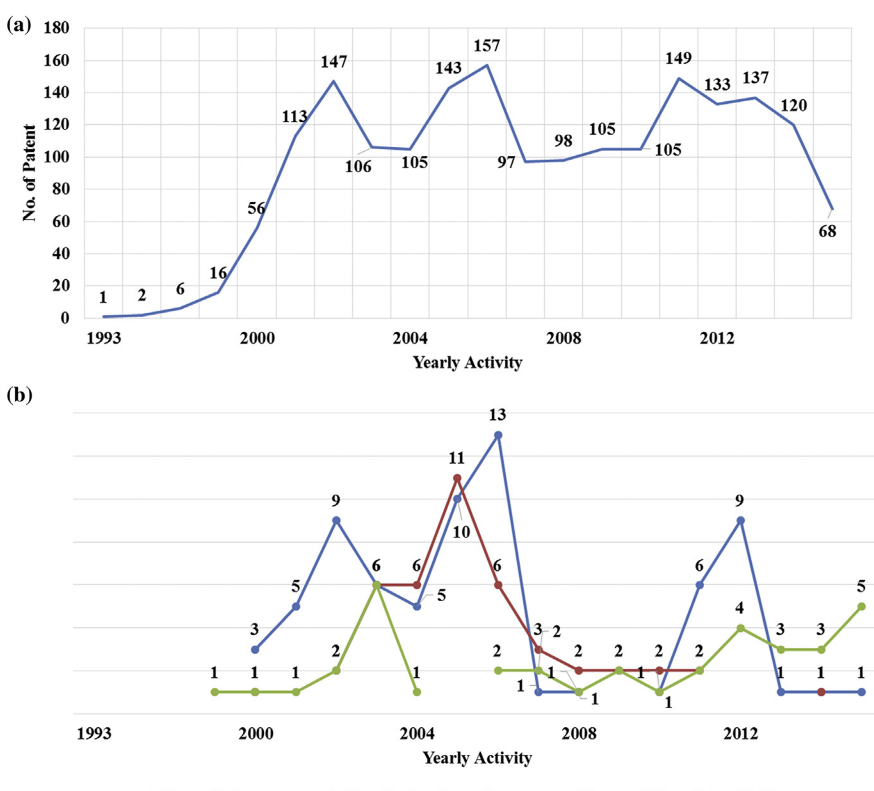


**Fig. 10.** (a). Patenting Trend in Hybridization or Gene Expression (HE). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Hybridization or Gene Expression (HE).

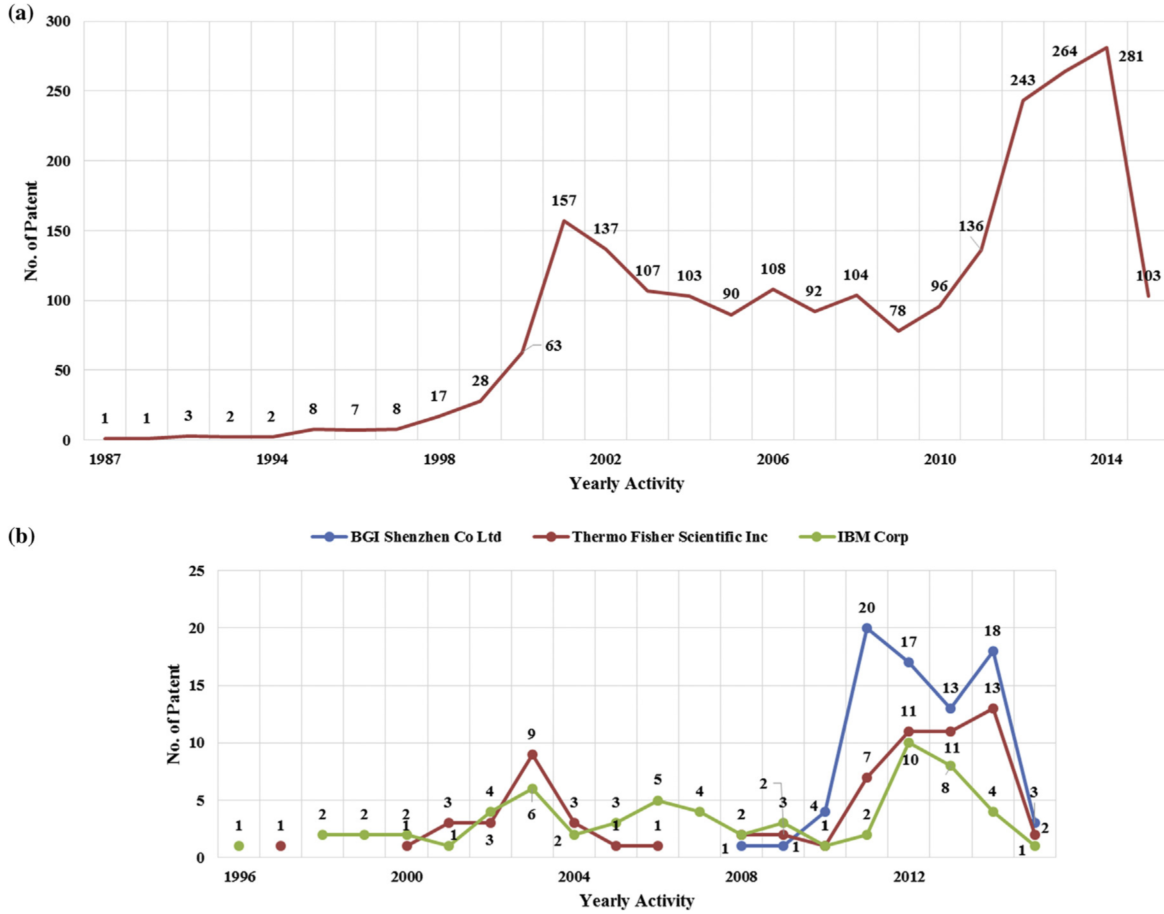


**Fig. 11.** (a). Patenting Trend in Sequence Comparison (SC). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Sequence Comparison (SC).

### 4.4.7. *Machine learning, data mining or biostatistics (G06F 19/24)*

In bioinformatics research, by using machine learning or datamining, significant information can be inferred from biological data such as to predict or analyze the diseases, any relationships associated with biological data, etc. [12]. Fig. 12 depicts patent filing trend for this technology. It is observed that activity increased from the year 2000 however, it is on decline in recent years (2014 and 2015). Fig. 12 reveals that in the years 2008–2012, the activity of Philips N.V. was highest but after the year 2012 Thermo Fisher Scientific Inc. and IBM Corp came into dominate the picture, although Philips N.V. remained in top position by holding a large number of patents/applications than others.

### 4.4.8. *Data visualization (G06F 19/26)*

It involves the application of computational and information methods which provide the insight into the complex study of biological data [13]. Fig. 13 represents drift in technology seen after the year 2000. From year 2011 a sudden increase in patents/applications number was observed. Fig. 13 reveals that Hitachi Ltd by filing large number of patents/applications between the years 2000–2006 was once a dominant player in this area. Whereas, Thermo Fisher Scientific Inc. is observed to be an active applicant in recent years whose number grew in 2014 and 2015.

### 4.4.9. *Programming tools or database systems (G06F 19/28)*

For management of valuable biological information, the huge amount of tools and databases have come up, which has strengthened the field of bioinformatics [14]. Fig. 14 reveals that although filing started in the 1970s, it peaked up in the year 2003. Fig. 14 indicates that Hitachi Ltd was the only active player in years 2000–2007 after which, Thermo Fisher Scientific Inc. has gained with a gradual growth in the patent filings in this technological area.

### 4.5. *Legal status scenario*

The bar graph (Fig. 15) represents the legal status. It reveals that the maximum number of patents/applications of Thermo Fishers Scientific Inc and its strong patent application pipeline. On the other hand, IBM Corp although, top active player in the list, has a less number of in-process applications.

## 5. Discussions

The landscape analysis was carried for the bioinformatics related patents. The patents/applications filing trend revealed that the field had emerged as a science of analyzing, interpreting, representing and sorting biological data over a period. It was observed that the patenting activity started in the 1970s. The patenting activity in bioinformatics started to rise from the year 2002, and the patenting activity reached a peak in the year 2013. Among the nine broadly classified technologies, Functional Genomics or Proteomics (GP) was observed to be foremost area with a large number of patents/applications filings followed by Sequence comparison (SC) and Machine Learning, Data Mining or Biostatistics (MDB). The assignee trends suggested that Thermo

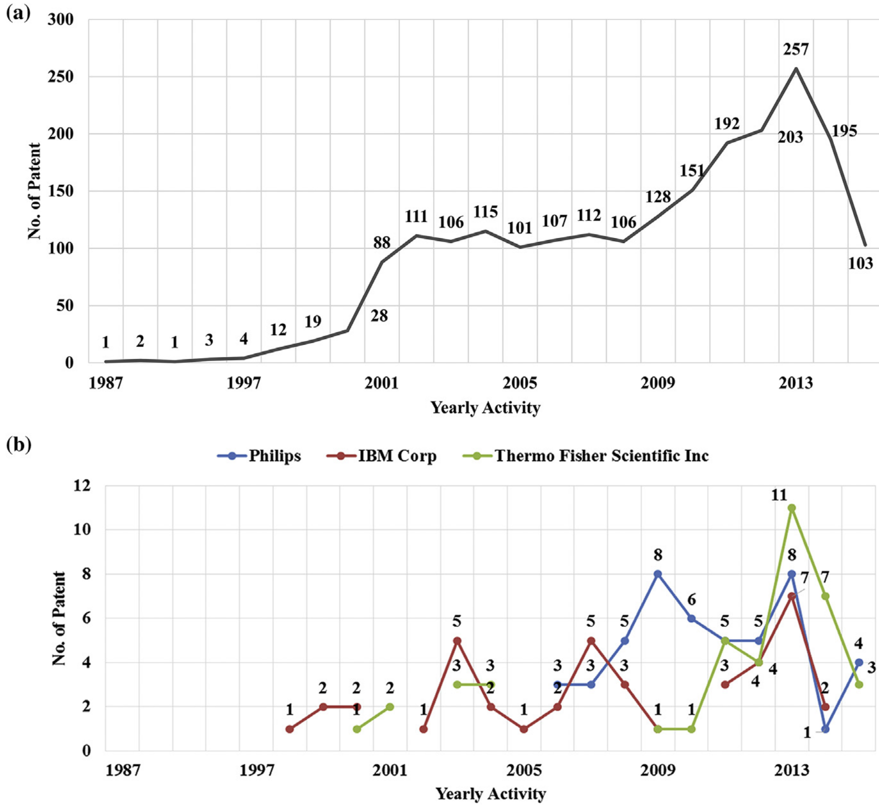


Fig. 12. (a). Patenting Trend in Machine Learning, Data Mining or Biostatistics (MDB). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Machine Learning, Data Mining or Biostatistics (MDB).

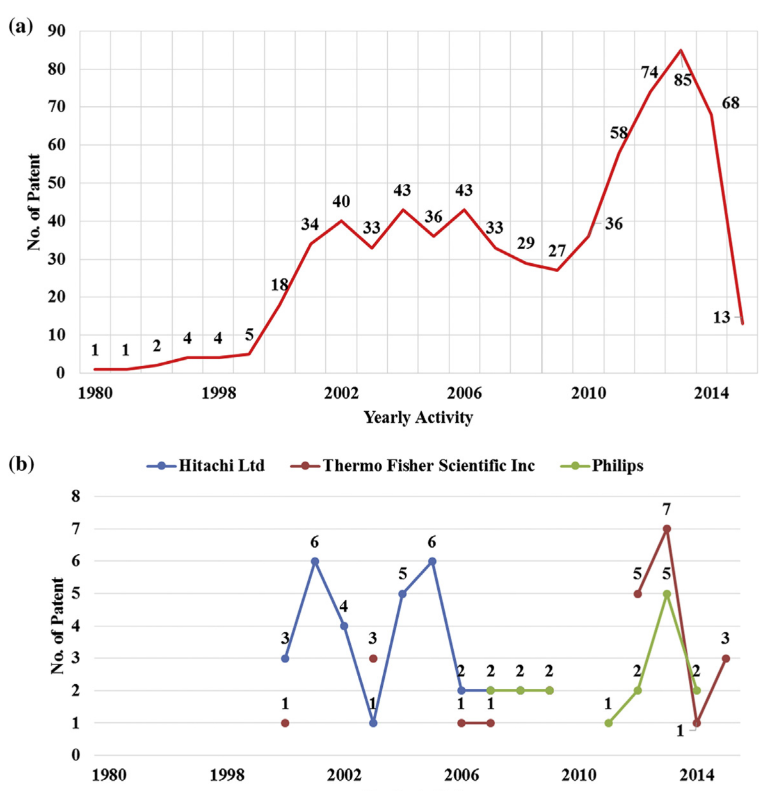


Fig. 13. (a). Patenting Trend in Data Visualization (DV). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Data Visualization (DV).

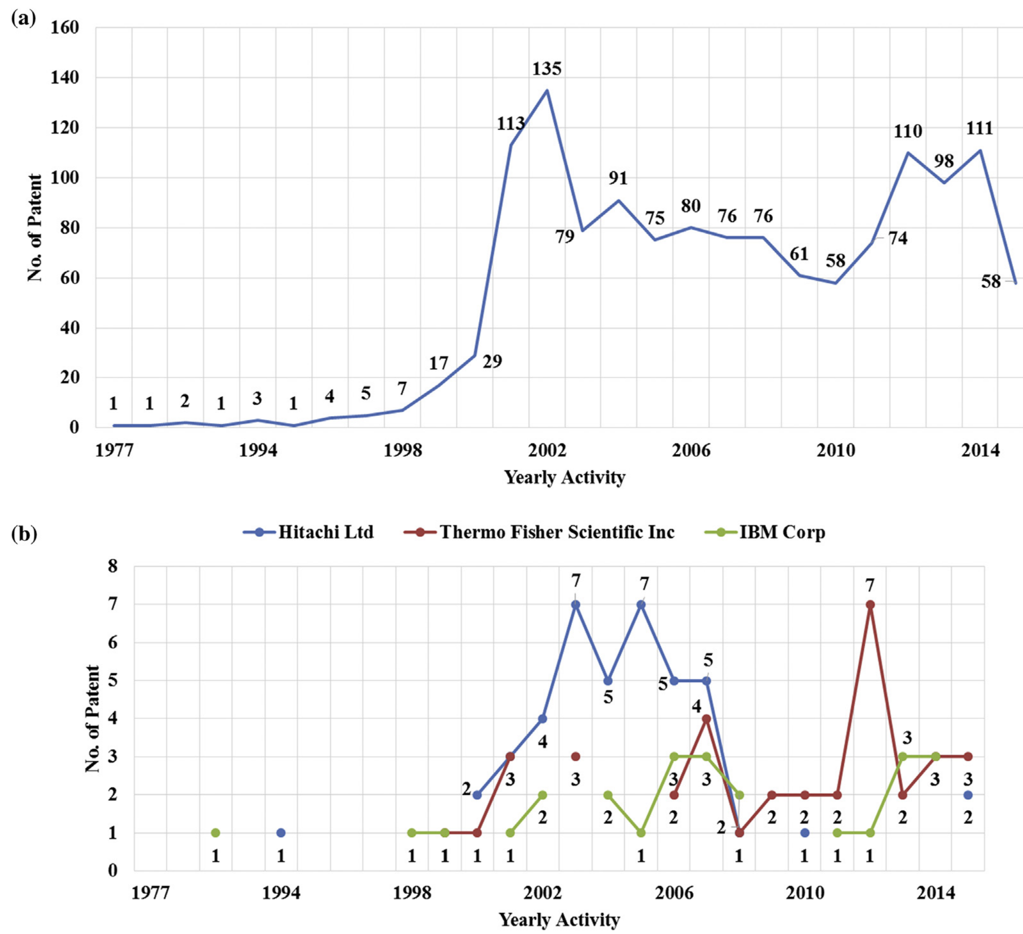


**Fig. 14.** (a). Patenting Trend in Programming Tools or Database System (PD). The overall patents/applications filing activity. (b). Top Assignee Patenting Trend. The view of major active players in Programming Tools or Database System (PD).

Fisher Scientific Inc. is the leading player (majorly in 2013) followed by Hitachi Ltd. Hitachi Ltd. had dominated in between the years (2002–2005) probably because in the year 2000, Hitachi Ltd. stepped into the biotechnology and bioinformatics area by establishing a Life Science Group [22]. However, in recent years, it was observed that Thermo Fisher Scientific Inc. surpassed

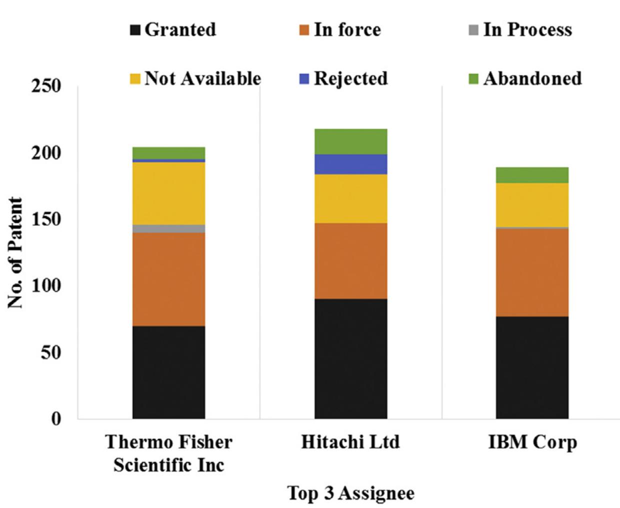


**Fig. 15.** Legal Status. The current legal status along with count for the filed patents/applications.

Hitachi Ltd. and had maximum fillings 2013 onwards. In the year 2013, Life Technologies Corp was acquired by Thermo Fisher Scientific Inc. which has created an unrivaled leader who serves research, diagnostics, and applied markets [23]. Superficially it appears that the top three assignees are from different background, however they all are related by being in areas related to diagnostics and that is the reason they have been actively filing bioinformatics related patents. Thermo Fisher Scientific is US based Multinational Corporation that is active in areas related to Analytical instruments, equipment, reagents and consumables. The corporation also provides software and services for research, analysis, discovery and diagnostics and may thus have filed patents in the domain of bioinformatics especially functional genomics/proteomics. Partly private and partly public, receiving funds from private investors as well as from Chinese government, BGI (Beijing Genomics Institute) is involved in genome sequencing activity. It also provides informatics services to various pharmaceutical companies that operate globally. Hence it is obvious that they file patents in this area. Philips is amongst the leaders in the area of diagnostics and has been investing in hiring of new talent, in building life sciences facilities aiming to build deep understanding of disciplines such as molecular biology, bioinformatics for applications in diagnostics which probably has led to them filing patents in this domain. Moreover, the country coverage analysis indicates that United States (US) had a maximum number of patents/applications filings followed by Japan and China.

## 6. Conclusion

This landscape study provided the patent situation (globally) in the area of bioinformatics by means of patent information analysis. Based on IPC and CPC classifications assigned to patents/application, the study categorizes the field of bioinformatics into nine subcategories. It was found that most of the research in the area on study of genes and protein expression is carried out in US followed by Japan. Also, limited number of patents have been files in tha area of phylogeny category.

This landscape study indicated the past and present patenting activities and trends in the bioinformatics field. Based on these trends and the insights derived from it, the researchers can plan and align their research activites or focus on plan to enter in the specified field in the technological area. The study also reveals the important players in the space, their opponents as well as the chronological growth in respective fields of technology.

We assume that in coming years, the field will continue to remain preferred source of biological knowledge as it will encompass an increasing flood of new data to be processed on computers of ever-increasing speed and storage space [24].

## Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.wpi.2017.11.007.

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